

Section A

Answer **all** the questions in this section.

- 1 Blackflies bite humans and suck blood. Blackfly larvae live in rivers and feed by filtering particles out of the water. A method of controlling blackfly is to add particles of an inactive toxin to the water upstream of where the larvae are feeding. The toxin is a protein.

A protease in the larva's digestive system activates the toxin. The active toxin then joins to a receptor molecule on the gut epithelial cell membranes. This damages the epithelial cells and kills the larva.

Fig. 1.1 shows the results of an experiment to determine the optimum pH for activation of the toxin.

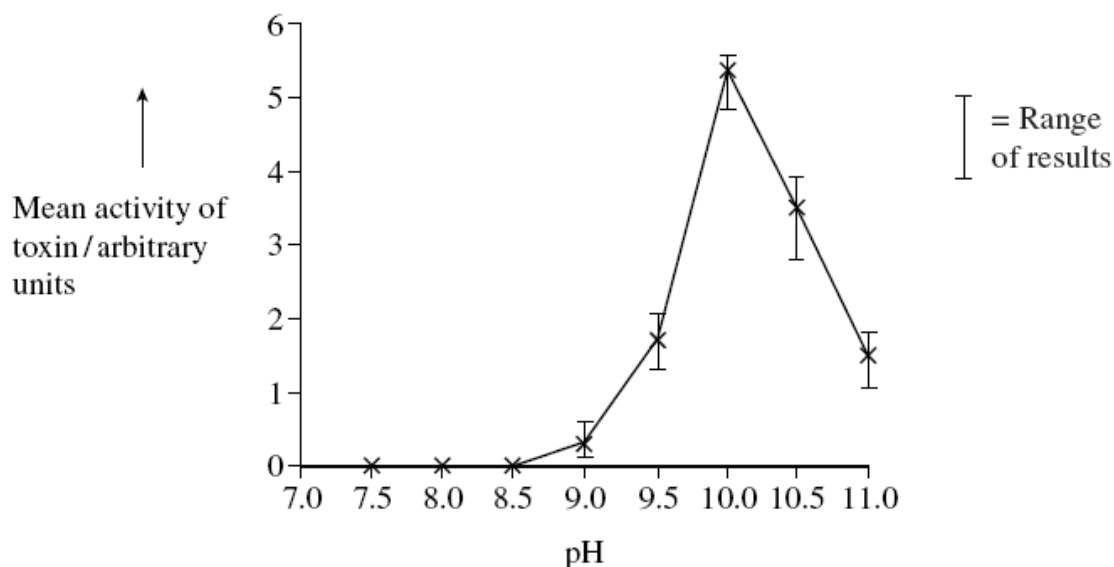


Fig. 1.1

- (a) (i) Using the data in Fig. 1.1, describe the effect of pH on mean activity of the toxin.

- Increasing pH from 8.5 to 10 increases mean activity of toxin from 0 to 5.5 a.u;
- Optimum pH at 10, highest mean activity of toxin at 5.5 a.u;
- Increasing pH from 10 to 11 increases mean activity from 5.5 to 1.5 a.u;

Any 2 [2]

- (ii) Explain why there is no activity of toxin at pH below 8.5.

- increase H^+ concentration;
- alters charges of R-groups of a.a ;
- of protease; reject toxin
- disrupts named bonds : ionic / hydrogen bonds;
- disrupts active site / tertiary structure;
- toxin (substrate) cannot bind to active site of enzyme/ ref to lock-key hypothesis / no formation of ES complexes ;

[4]

(iii) Suggest how the protease in larva's digestive system activates the toxin.

protease cleave/ hydrolyse part of amino acid chain of toxin;
toxin has the correct configuration to bind to receptor molecule;

[2]

(b) Part of the blackfly larva's gut has a pH of 10 to 10.5. All other animals living in the river have gut pH values below 8.5.

Suggest why the toxin can be used for selective biological control of the blackfly.

- High pH needed to activate toxin found only in blackfly/ toxin inactive at or <pH 8.5/ high pH needed for protease activity
- Specific protease needed to activate toxin found only in blackfly
- Specific receptor found only in blackfly;

Any 1 [1]

(c) Phosphofructokinase (PFK) is an important enzyme which is thought to control the rate of glycolysis. The graph in Fig. 1.2 shows PFK activity at increasing substrate concentration at two levels of ATP.

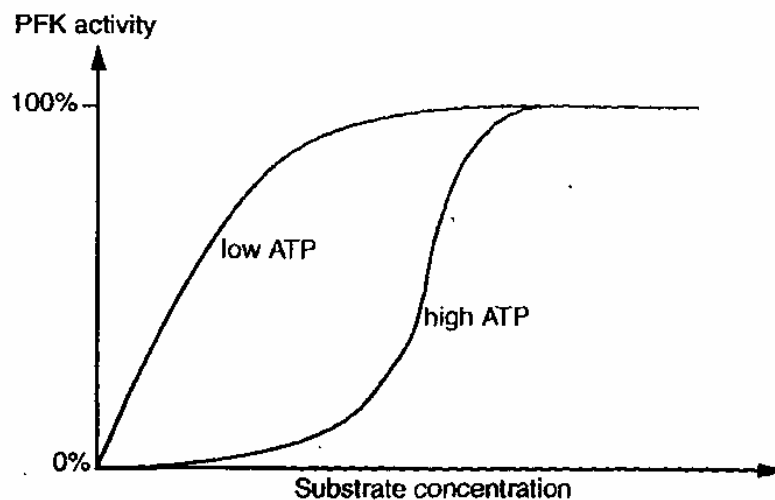


Fig. 1.2

(i) With reference to the figure above, describe the effect of high levels of ATP on PFK activity.

enzyme inhibited/decrease at high levels of ATP

[1]

(ii) Explain how the rate of glycolysis may be controlled by this enzyme.

High levels of ATP causes negative feedback;

ATP is allosteric inhibitor of PFK / binding of ATP to allosteric site of PFK;

causing change in shape of active site of PFK/ no binding of substrate (fructose-6-phosphate) to active site;

decreases rate of glycolysis;

Accept converse argument (Low levels of ATP promotes PFK activity so increases the rate of glycolysis;.....)

[2]

[Total : 12]

- 2 Fig. 2.1 shows the polycistronic nature of *lac* operon in *E.coli*.

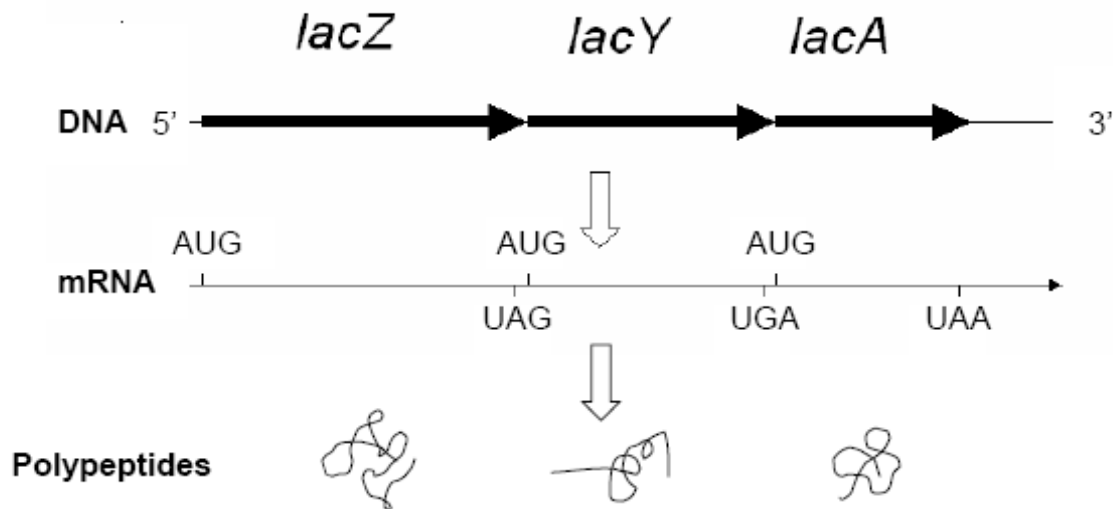


Fig. 2.1

- (a) With reference to Fig. 2.1, explain why the *lac* operon is polycistronic in nature

mRNA of the *lac* operon contains three start codons and three stop codons.

three different polypeptides can be translated from / encoded by one mRNA

[2]

A certain bacteriophage is known to insert its own genome within the *lacZ* gene of *E.coli*. In an experiment, *E.coli* cells were infected with the bacteriophages and the infected cells grown in a medium containing equal amounts of glucose and lactose.

Fig. 2.2 shows how the growth of infected *E.coli* cells varies with concentration of glucose and lactose.

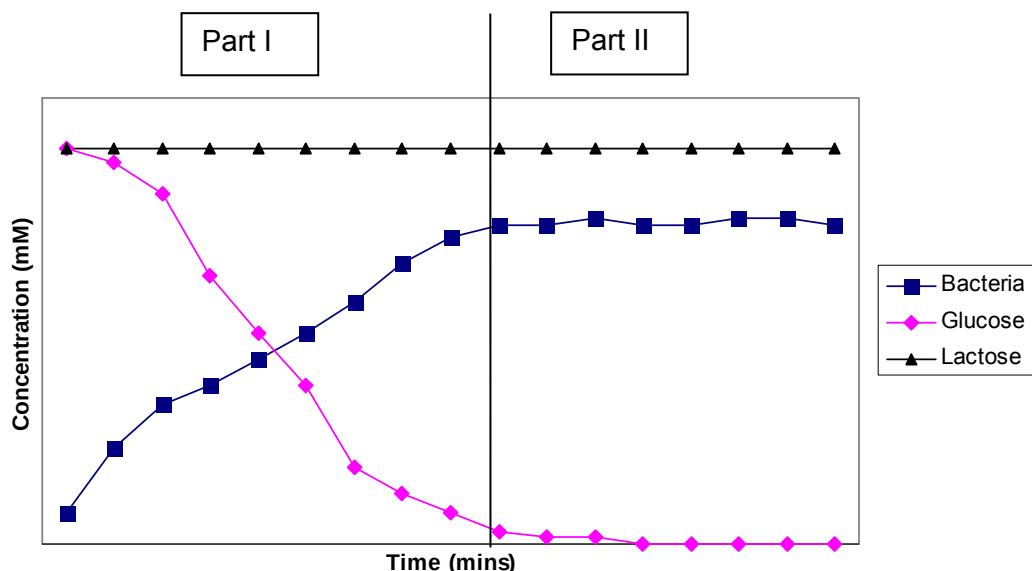


Fig 2.2

- (b) With reference to Fig. 2.2, explain

- (i) **Part I** of the graphs;

increase in number of *E.coli* as glucose concentration decrease;

glucose is preferred respiratory substrate;

no change in concentration of lactose as *lac* operon not expressed;

[2]

(ii) **Part II** of the graphs

growth of *E.coli* plateaus/ constant because glucose depleted / concentration decrease to zero;

lactose concentration is still constant;

because *lac Z* gene disrupted hence no β -galactosidase produced;

[2]

- (c) To determine the number of infectious bacteriophages, we can exploit the fact that viral infections can result in the lysis of the bacterial host cell which shows up as a plaque or a zone of clearing on an agar plate containing a heavy growth of bacteria.

Bacteriophages can thus be quantified by means of a plaque assay. The term 'plaque forming unit (PFU)' gives a measurement of the number of infectious viral particles present during an experiment.

Fig. 2.2 below shows the changes in PFU caused by temperate bacteriophages infecting *E. coli* cells.

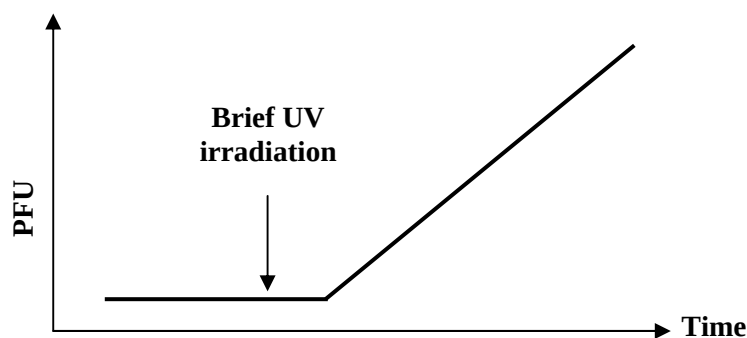


Fig. 2.2

- (i) Suggest a possible explanation for the flat part of the graph as shown above.

Bacteriophages undergoing lysogenic cycle / not making new viral components

[1]

- (ii) Describe how bacteriophages infecting the *E. coli* cells resulted in an increase in PFU shortly after UV irradiation.

UV triggers switch from lysogenic cycle to lytic cycle / induction of lytic cycle;

prophage excised from host (bacterial) chromosomes;

Enzymes in *E. coli* host cells replicate viral DNA/ transcribe and translate the viral DNA into new viral components;

Viral DNA and viral proteins are assembled into new viral particles;

Lysis of host cells to release bacteriophages;
to infect other host cells;

[4]

- (iii) Explain the significance of the ability of temperate bacteriophages to co-exist with their bacterial host cells without causing lysis of the host cells.

So that host cell can reproduce the viral DNA / prophage in all host cells' offspring as the host cells divide by binary fission

[1]

[Total: 12]

- 3 Fig. 3.1 shows the binding of a general transcription factor to part of the promoter.

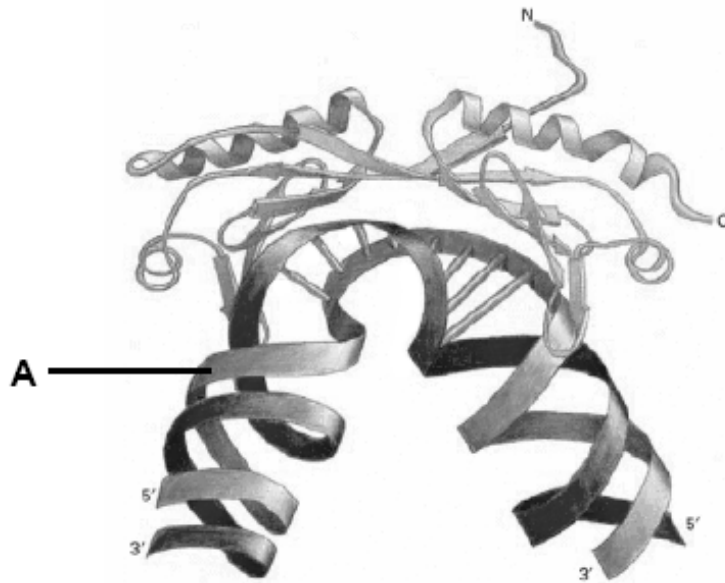


Fig 3.1

- (a) (i) With reference to Fig. 3.1, give **two** reasons why the structure labeled **A** is part of the promoter and not the transcription factor.

- Promoter is made of DNA;
- double helix / stranded;
- Has 5' and 3' ends;
- Antiparallel

[2]

- (ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter.

- New/different sequence → different shape
- not complementary to binding site of / not recognised by transcription factors / RNA polymerase;
- Transcription factors / RNA polymerase cannot bind;
- Transcription cannot be initiated;

[2]

- (b) An *in vitro* transcription under the control of a eukaryotic promoter has been developed. The transcription of DNA can occur in

Nuclear extract: with contents of cell nucleus without genome

Purified system: with purified RNA polymerase with the following transcription factors, TFIID (the TATA binding factor), TFIIB and TFIIE (which bind to RNA polymerase).

A series of experiments that compare the efficiency of transcription in the purified system with the efficiency of transcription in a crude nuclear extract have been performed. The two systems were tested with the undeleted template DNA and with various deleted templates that were generated as shown in Fig. 3.2. The number represents the number of bases away from the transcription start site (+1).

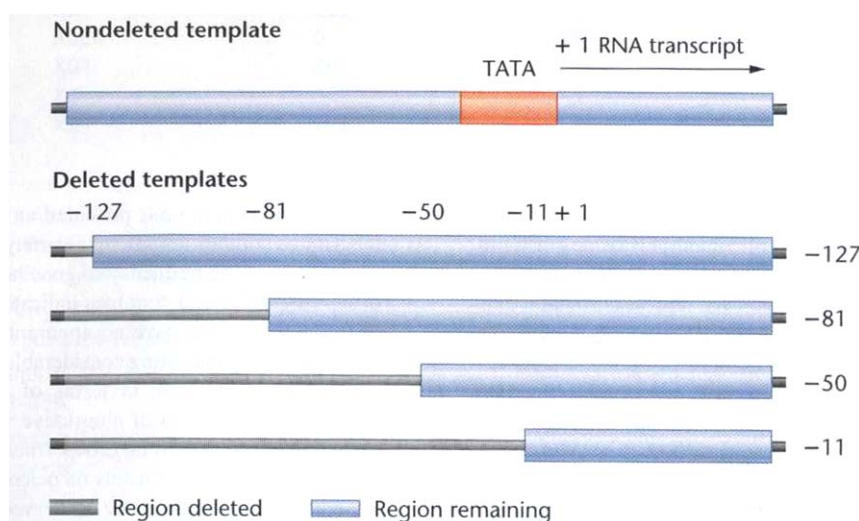


Fig. 3.2

Results of the test are shown in Table 3.2.

DNA template added	Nuclear extract	Purified system
undeleted	++++	+
-127 deletion	++++	+
-81 deletion	++++	+
-50 deletion	+	+
-11 deletion	o	o

Table 3.3

Legend

- + Low efficiency transcription
 ++++ High efficiency transcription
 o No transcription

- (i) Explain why there is no transcription from -11 deletion template.

- Missing promoter sequence/ missing TATA box
- RNA polymerase/ transcription complex cannot assemble

[2]

With reference to Table 3.3,

(ii) explain the difference in the results between the nuclear extract and the purified system for the undeleted template.

- Nuclear extract has higher rate of transcription
- Nuclear extract has more/all/ more specific transcription factors needed for high rate of transcription
- Higher stability of transcription complex

[2]

(ii) explain the difference in the results between the -50 deleted template and -81 deleted template in the nuclear extract.

- -50 deleted template has basal/ lower transcription rate
- Contained only the sequence for RNA polymerase and 2 basal/general transcription factors to bind/ DNA sequence for the rest of the transcription factors to bind have been deleted/ are beyond -50

[2]

(c) DNA methylation is commonly associated with reduction of transcription. Irvine and colleagues (2002. *Mol. And Cell. Biol.* 22:6689-96) studied the impact of location of DNA methylation relative to gene activity in human cells.

Table 3.4 below describes the relative expression of a reported gene (luciferase) with variable DNA methylation outside and within the transcription region.

DNA Segment	Patch Size of Methylation (kb)	Number of Methylated CpGs	Relative Luciferase Expression
Outside transcription unit (0 -7.6kb away)	0.0	0	490X
	2.0	100	290X
	3.1	102	250X
	12.1	593	2X
Inside transcription unit	0.0	0	490X
	1.9	108	80X
	2.4	134	5X
	12.1	593	2X

Legend: CpG –CG-rich DNA regions

Table 3.4

With reference to Table 3.4, describe the effects of DNA methylation on luciferase gene expression.

- Reduction in luciferase expression is proportional to the amount of methylation and patch size. (Quote)

- Methylation within the transcription unit reduces the luciferase expression more drastically than the methylation outside the transcription unit. (Quote)
- A high degree of methylation outside the transcription unit(593CpGs) has as great a impact on depressing transcription as the same degree of methylation within the transcription unit. (Quote)

[2]

- (d) Describe one other transcriptional control that can help increase the rate of transcription of a gene in a cell?

chromatin modification: histone acetylation adds negative charge to histones, decreasing interaction of histones with negatively-charged DNA

Activators bind to distal control elements / enhancer;
(More) transcription factors and RNA polymerase brought to promoter;

[2]

[Total: 14]

- 4 In corn, coloured aleurone (in the kernels) is due to the dominant allele **R**. The recessive allele **r**, when homozygous, produces colourless aleurone. The plant colour (not the kernel colour) is controlled by another gene with two alleles, **P** and **p**. The dominant **P** allele results in purple colour, whereas the homozygous presence of the recessive **p** allele causes the plant to appear yellow.

In a testcross between a plant of unknown genotype and a plant that is homozygous recessive for both traits, the following progeny were obtained:

Coloured purple	88
Coloured yellow	12
Colourless purple	8
Colourless yellow	92

- (a) (i) Draw a genetic diagram for the above cross

[4]

1m for parental phenotype, coloured green and colourless yellow (x in between)

1m for parental genotype, RrGg and rrgg (x in between)

1m for gametes, RG, rg, Rg, rG and rg

1m for F1 genotype with correct phenotype

- (ii) Explain the results of the testcross

1m for indicating recombinant genotype/ parental genotype (award if student indicated in (i))

1m stating gene R and gene G are linked/ found on the same chromosome

[2]

- (b) Recombination frequency is defined as the frequency that a chromosome crossover will take place between 2 gene loci during meiosis. The recombinant frequency is calculated by the following method:

$$[\text{Total number of recombinants} / \text{Total number of progeny}] \times 100\%$$

- (i) Calculate the recombinant frequency between gene of aleurone colour and gene of plant colour. Show your working.

$$[20 / 200] \times 100\% = 10 \%$$

[1]

- (ii) The recombinant frequency also represents the map units between the genes.

Explain how recombination frequency can be used to determine the distance between 2 gene loci.

Higher frequency , gene loci further apart / lower frequency gene loci closer together

[1]

- (b) An old form of wheat (*Triticum turgidum*), has a diploid chromosome number of 28 ($2n = 28$). A wild wheat, einkorn wheat (*Triticum tauschii*), has a diploid chromosome number of 14 ($2n = 14$).

These two species occasionally crossed and produced sterile hybrid plants. Due to an error during cell division, one of these hybrid plants formed male and female gametes with 21 chromosomes. Fusion of these gametes results in viable offspring. These plants were a new species, *Triticum aestivum* ($2n = 42$), our modern bread wheat.

- (i) How many chromosomes would there have been in each of the cells of the hybrid plant produced by crossing *Triticum turgidum* with *Triticum tauschii*?

21

[1]

- (ii) State the type of cell division in which the error occurred.

Mitosis;
Reject meiosis

[1]

- (iii) Suggest what error occurred during cell division that enabled the sterile hybrid plants to produce gametes.

Chromosome doubling/ somatic doubling/ non-disjunction of entire set of chromosomes/ no separation of sister chromatids;

Spindle failure;

No cytokinesis;

[1]

max 1

- (iv) Explain why *Triticum aestivum* is fertile while majority of hybrid plants were not.

1. *T. aestivum* has 2 copies of each type of chromosome/is diploid;
 2. *T. aestivum*'s chromosomes can form bivalents/can assort in meiosis/ can produce haploid gametes;
 3. *T. aestivum*'s gametes receive a copy of every chromosome/ receive all the genetic information;
- [ACCEPT converse argument for hybrid plants]

[3]

[Total: 14]

- 5 In disorders such as retinoblastoma, a mutation in one allele of the RB1 gene can be inherited from the germline, causing an autosomal dominant predisposition to the development of eye tumours. To develop tumours, a somatic mutation in the second copy of the RB1 gene is necessary, indicating that the mutation itself acts a recessive trait.

(a) Given that the first mutation can be inherited, describe **one** way in which a second mutational event can occur.

1. nucleotide alteration: nucleotide/base substitution/ deletion/ insertion
2. loss of chromosome (13) during mitosis → chromosome monosomy
3. loss of chromosome (13) followed by duplication of chromosome (carrying mutant allele)
4. deletion of long arm/ chromosomal deletion
5. interstitial deletion involving the RB1 locus

[1]

Ras gene is an example of proto-oncogene. The ras gene product is a signal transduction protein that operates as an on/off switch in response to external stimulation in a cell-cycle stimulating pathway. The cellular response at the end of the pathway is the synthesis of a protein that stimulates the cell cycle.

Fig. 5.1 show the sequence of events in the cell-cycle stimulating pathway.

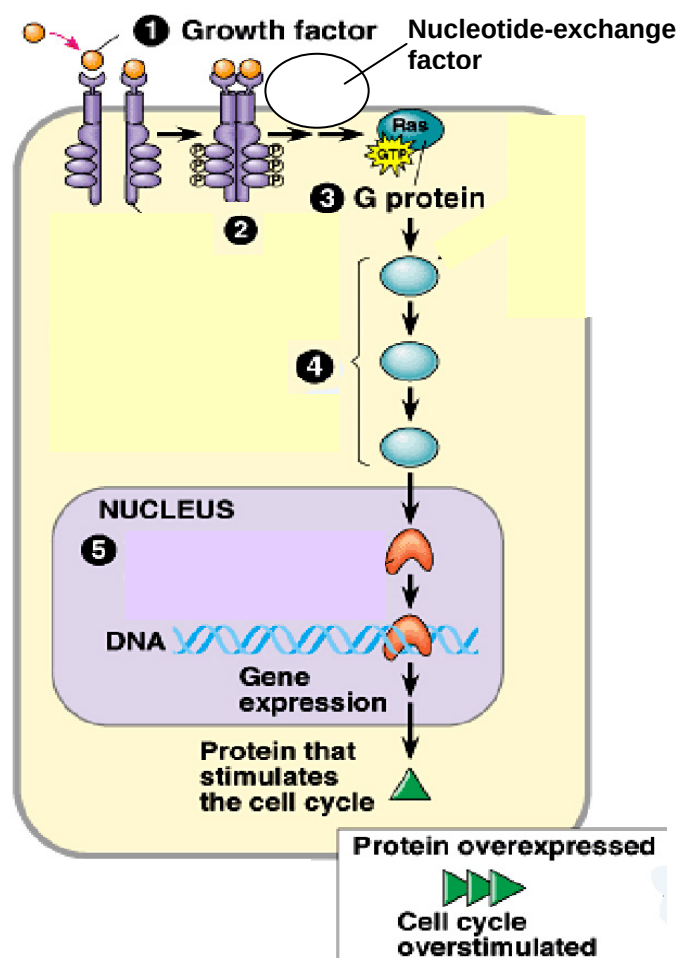


Fig 5.1

- (b) With reference to Fig. 5.1, outline the cell-cycle stimulating pathway that led to the synthesis of a protein that stimulates the cell cycle.

1. growth factor binds receptor;
2. on CSM;
3. causing dimerisation of receptors;
4. phosphorylation (of intracellular tails/receptors);
5. ras → G protein;
6. recruitment of nucleotide exchange factor/protein;
7. GTP replaces GDP;
8. Ras becomes active
9. Activates phosphorylation cascade/ series of relay proteins;
10. Activate transcription factors
11. Gene expression/ transcription & translation;

(½m each)

[4]

- (c) Some retrovirus can cause cancer in animals. When a retrovirus that does not carry an oncogene infects a particular cell, northern blots indicate that the amount of mRNAs transcribed from a particular proto-oncogene became elevated approximately 13-fold compared with uninfected control cells.

Suggest an explanation for this.

→ viral DNA integrates near the proto-oncogene;

[2]

viral promoter/ enhancers stimulate high level of transcription of the proto-oncogene/ translocation (strong LTR);

Mutation but substantiate with explanation (translocation/ gene amplification);

- (d) Suggest why many oncogenic viruses contain genes whose products interact with tumour suppressor proteins

Why viruses need hosts →

Virus need host cell to be in actively dividing state in order to access the host DNA synthetic enzymes/ host cell needs to undergo cell division;

Suggest HOW virus stops host →

Uncontrolled cell division/ absence of apoptosis/ absence of DNA repair → cell proliferation
Any 2

[2]

[Total : 9]

- 6 Fig. 6.1 shows the relationship between the diameter of the axon and the speed of conduction of nerve impulses in myelinated axons of a cat.

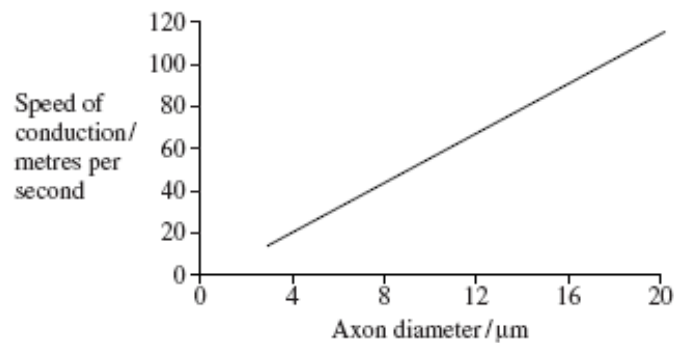


Fig. 6.1

- (a) (i) With reference to Fig. 6.1, explain the relationship between axon diameter and speed of conduction.

As axon diameter increases, speed of conduction increase (quote values with correct units)

Increase diameter, decrease resistance of axoplasm/ increase spread of current/ increase length of local circuits [2]

- (ii) Describe how an action potential is propagated along a myelinated neuron.

Voltage-gated Na^+ channels open/ Na^+ influx;
depolarise region at node of Ranvier; **REJECT: REGION**
Positive feedback/ trigger more voltage-gated Na^+ channels open/ further Na^+ influx; **bulk of cohort missed this point**
Membrane potential increase above threshold potential $\rightarrow$ trigger AP; [2]
($\frac{1}{2}$ m each)

- (iii) Explain why a myelinated axon uses less ATP to transmit a nerve impulse than an unmyelinated axon of the same diameter.

2 parts:

1) compare myelinated **vs** non-myelinated

2) what is the ATP used for (most did not address this part)

ATP required for active transport by Na^+-K^+ pump; [2]
 Na^+ (actively) moved out only at nodes in myelinated / Na^+ (actively) moved out
along whole length of axon in non-myelinated during restoration of action potential;

Voltage-gated sodium channels are transmembrane proteins that produce the ionic current responsible for the rising phase of the muscle action potential.

Paramyotonia congenita is a genetic disease that involves muscle stiffness and weakness, especially in the face and upper extremities. One of the causes of paramyotonia congenita is a single-base substitution in a subunit of the sodium channel. The mutant protein is full-length and permits sodium entry like a normal channel would but it is unable to close as quickly as a normal sodium channel.

(b) Fig. 6.2 shows a normal action potential.

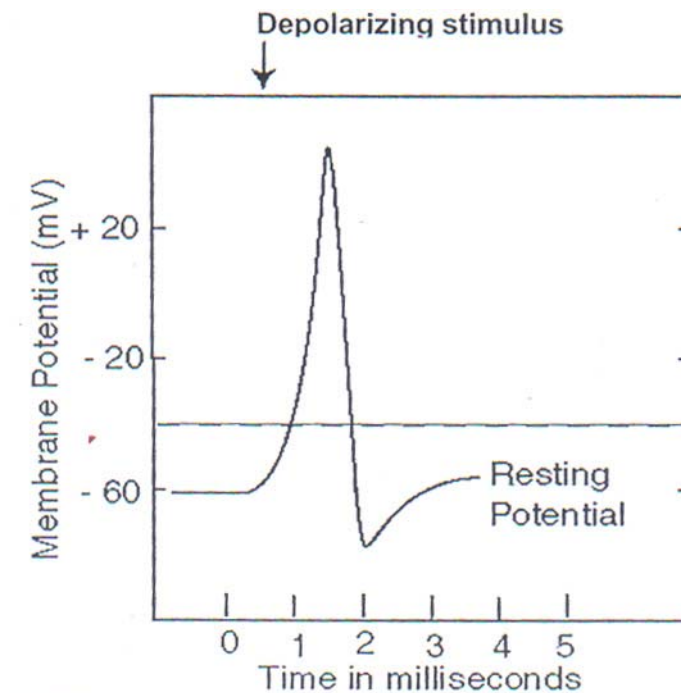


Fig. 6.2

- (i) Sketch on Fig. 6.2 above, how you would expect the action potential graph for a sufferer of paramyotonia congenita to differ.

$\frac{1}{2}$ m for spike above normal magnitude/duration

$\frac{1}{2}$ m for a slower, more eventual return to RMP

[1]

- (ii) Explain your sketch.

Increased time for sodium influx $\rightarrow$ increased spike magnitude/duration;

However since potassium channels are still functional and sodium channels eventually $\rightarrow$ repolarisation still occurs but slower;

Hyperpolarisation is less/non-existence as the amount of potassium entering is the same [2]

Any 3 points

[Total: 9]

- 7 Fig. 7.1 shows the classification of some primates.

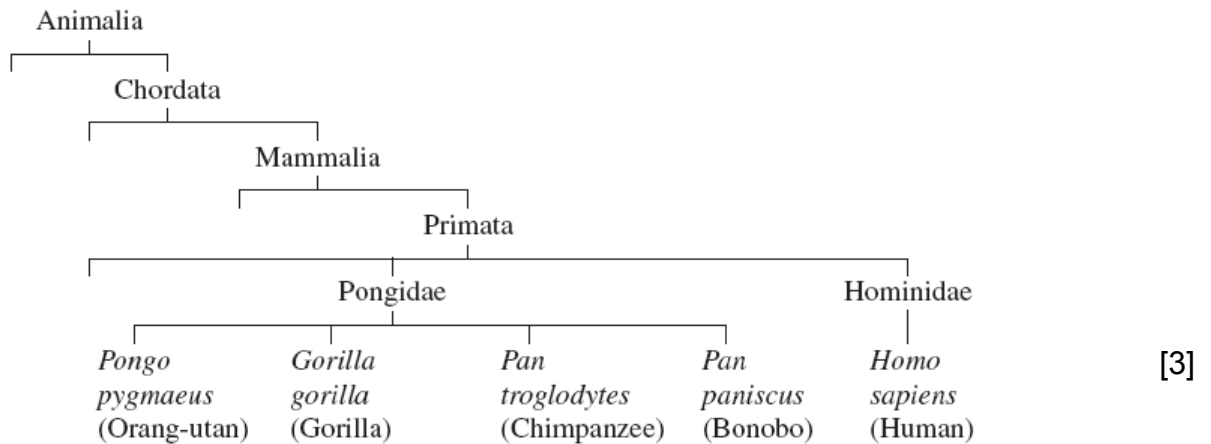


Fig. 7.1

- (a) This classification is hierarchical and phylogenetic. Explain what is meant by

- (i) hierarchical

(Large) groups are divided into smaller groups;

[1]

- (ii) phylogenetic

Based on evolutionary history / linked to common ancestors;

[1]

- (b) Many years ago, scientists used a different system of classification of the same primates. This is shown in Fig. 7.2.

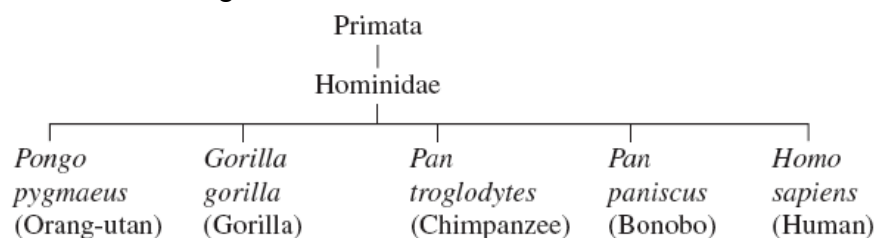


Fig. 7.2

- (i) Using information from Fig. 7.1 and Fig. 7.2, give **one** way in which the classification of these primates has changed.

Logical comparison between the FAMILIES (either using Human/ or Orang-utan, Gorilla, Chimpanzee, Bonobo);
Or if unable to state families, mention BOTH Hominidae and Pongidae;

[1]

- (ii) Suggest **one** new source of evidence that has contributed to the changes in classification.

DNA hybridization/ DNA / base sequences/ amino acid sequences (in proteins)/ fossils/ Immunology/ embryology;

1 max

[1]

Zooplankton are very small animals which feed on algae (green protocists) found in lakes.

Fig 7.1 shows the mean depth of zooplankton populations in four lakes, P to S, during the day and the night. It also shows the standard deviations of the means. The depth at which the light intensity is 1% of the surface light intensity is also shown.

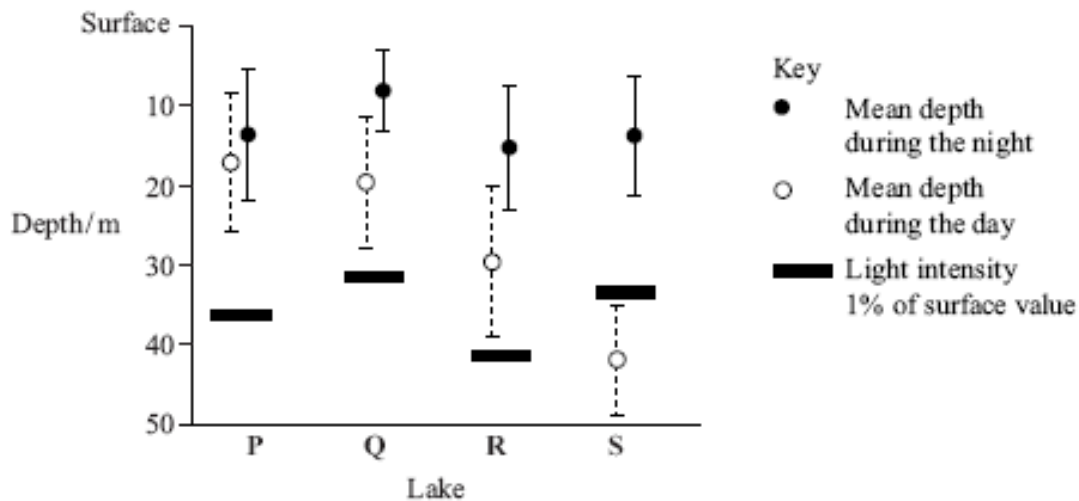


Fig 7.1

- (c) Explain how Fig 7.1 provides evidence that the zooplankton feed at night.

For all lakes, zooplankton are nearer to the surface at night because algae only found at surface;

algae undergoes photosynthesis since there is no/little light below 30/40m;

Predatory fish, which hunt by sight, are present in some of the lakes. Table 7.1 shows the different lengths of time in which predatory fish have been present in the lakes.

Lake	Estimated length of time predatory fish have been present/ years
P	0
Q	5
R	25
S	Over 1000

Table 7.1

- (i) Describe the relationship between the depth of the zooplankton during the day and the length of time predators have been present in the lake.

The greater/deeper the depth at which the zooplankton occur during the day, the longer the time predators have been present in the lake

- (ii) Suggest how the differences in behaviour of the zooplankton populations in the four lakes might have evolved.

variation in migration behaviour;

vertical migration reduces chance of predation/prey can't be seen in

low light intensity/ those with ability to migrate to lower depth are at selective advantage;

more likely to survive and reproduce to pass down genes/alleles (for behaviour) to next generation;

increase in frequency of gene/allele in population;

[Total:10]

8 (a) Compare the molecular structure of glycogen and triglycerides

[6]

Similarity:

- both are made up of **carbon, oxygen and hydrogen**
- both are **covalently bonded**
- both have **high % of hydrocarbons (C-H)**

Difference:

Glycogen	Triglyceride
Monomers are made up of glucose	Monomers are made up of glycerol & fatty acid chains
Bonds linking up the monomers are glycosidic linkages	Bonds linking up the monomers are ester linkages
	Has more hydrogen atoms for the same number of carbon
	Has more C-H bonds for the same mass

(b) Discuss the advantages to organisms of storing triglycerides rather than glycogen

[6]

Effective source of energy

- for the same weight, triglycerides have a higher energy value than carbohydrate (glycogen)
- 1g of triglycerides yields 38kJ energy, 1 g of carbohydrates yields 17kJ energy

Effective long-term storage of energy

- triglycerides is oxidized only after carbohydrates (glycogen) is depleted
- allows hibernating animals to store large amount of triglyceride as energy source which will be mobilised when carbohydrate is depleted.
- For the same amount of energy stored, mass of triglyceride is half of that of carbohydrate/ takes up less space than carbohydrate.
- allows seeds of plants to be kept small for ease of dispersal /minimises the mass of animal for greater freedom in locomotion.
- triglyceride is insoluble in water
- can be stored in large amount without affecting water potential of cells. This

Effect source of water

- triglycerides release more/ twice as much metabolic water as carbohydrates when oxidized

Others

- insulating material/ protect delicate organs/ provide bounyance/ electrical insulator (myelin sheath) for neurons – speeds up impulse transmission

(c) Explain the role of membranes in photosynthesis and respiration.

[8]

in photosynthesis

- thylakoid membrane
- site of light dependent reactions
- holds photosynthetic pigments/ PSI and PSII/ accessory pigments/ chlorophyll b and carotenoids in position
- for maximum absorption of light
- has primary pigments/ reaction centre/ chlorophyll a P680 & P700
- use energy from accessory pigments to excite/boost one of their electrons to a higher energy level

- transfer excited electron to primary electron acceptor.

in respiration

- inner membrane of mitochondria
- site of oxidative phosphorylation

in both processes

- inner mitochondria membrane highly folded/ form cristae
- thylakoid membrane organized into grana
- to increase surface area
- for holding e-carriers/ photosynthetic pigments/ stalked particles
- relatively impermeable
- to allow accumulation of protons/ generation of proton gradient
- hold e-carriers of ETC in decreasing energy level
- to allow energy to be released during electron passage along ETC
- hold proton pumps
- to harness energy loss during energy transfer to pump H⁺
- hold stalked particles / ATP synthase
- to allow flow of H⁺ back into stroma/matrix to provide energy for
- phosphorylating ADP to give ATP

@ ½ m each

- 9 (a) Compare the behaviour of chromosomes during mitosis and meiosis.

[6]

	Mitosis	Meiosis
Prophase	Synapsis does not occur. Homologous chromosomes remain separated	Prophase I: Synapsis occurs/ homologous chromosomes pair up
	No crossing over/ no chisama/ chiasmata formation	Crossing over between non-sister chromatids/ chiasmata and chisama formation occurs
Metaphase	Chromosomes align end to end at metaphase plate, independent of each other	Metaphase I: Homologous chromosomes align at metaphase plate, with one chromosome of each pair facing each pole. Metaphase II: Chromosomes align end to end at metaphase plate, independent of each other (<i>similarities</i>)
Anaphase	Identical sister chromatids separate/ centromeres divide	Anaphase I : Homologous chromosomes separate; Centromeres remain joined Anaphase II: Non-identical sister chromatids separate; Centromeres divide (<i>similarities</i>)

Similarities:

In both processes, chromatin fibers starts to condense into discrete chromosomes during early prophase

In both processes, chromosomes becomes less uncondensed in telophase

max 5 marks if no similarities mentioned

(b) Discuss the roles of RNAs in translation.

[8]

refer to summary notes of A4

(c) Describe the various ways in which the activity of an enzyme may be controlled after a mature mRNA that codes for it has been produced in a eukaryotic cell.

[6]

- Controlling mRNA stability through degradation by exonucleases
- Shortening of poly-A-tail to a certain length causes 5' cap to be removed
- and thus mRNA to be very susceptible to exonucleases

- Binding of micro RNAs (miRNA) complementary to mature mRNA
- will form double stranded RNA
- thus preventing ribosome from binding or moving down mRNA / translating mRNA
- or bring about the degradation of the mRNA

- Binding of regulatory proteins to untranslated region (UTR) on mRNA can also activate or inhibit translation
- by affecting binding of ribosome

- Degradation of synthesized proteins by ubiquitin
- determines length of time enzyme is active

- Modification of polypeptides to produce functional enzymes,
- . eg. glycosylation, dimerisation, folding into specific 3D configuration
- Action of inhibitors
- competitive and non-competitive.

End-product inhibition – allosteric

[Total: 20]